SAT Calculator Program — Complete Guide

TI-Nspire Toolkit · IvyTutoring.net · Practice Questions & Explanations for Every Function

How to use this guide: Each tool shows what inputs to enter, what the tool computes, and a practice SAT-style question you can try. Open the calculator on your TI-Nspire, navigate to the correct menu, and enter the inputs as shown. To print as PDF: File → Print → Save as PDF.

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1. Algebra (35% of SAT Math)

Linear equations, systems, inequalities, slopes, and line equations — the largest portion of the SAT.

Solve any Equation

Solves any single-variable equation for x. Shows steps: rearrange → factor → solve → verify.

Input: `Equation= 2x+3=7`

PRACTICE QUESTION

If 3(x – 2) + 5 = 2x + 7, what is the value of x?

Enter: `3(x−2)+5=2x+7`

Answer: x = 8

Solve 2×2 System

Solves a system of two linear equations using substitution and verification.

Inputs: `1.Eqn= x+y=5` `2.Eqn= 2x−y=1`

PRACTICE QUESTION

A store sells pencils for $0.50 and pens for $1.25. You buy 8 items totaling $6.50. How many of each?

Enter: x+y=8 and `0.5x+1.25y=6.5`

Answer: x = 4 pencils, y = 4 pens

Solve 3×3 System

Solves a system of three linear equations in x, y, z.

Inputs: `x+y+z=6` , `2x−y+z=3` , `x+2y−z=3`

PRACTICE QUESTION

If x + y + z = 10, 2x + y = 7, and y + 3z = 15, find x, y, and z.

Answer: x = 1, y = 5, z = 4

Linear Inequality

Solves linear inequalities in one variable using the CAS engine.

Input: `Inequality= 2x+3>7`

PRACTICE QUESTION

A phone plan charges $30/month plus $0.10 per text. If you want to spend less than $50, how many texts can you send?

Enter: `30+0.1x<50`

Answer: x < 200 texts

Simplify & Evaluate

Expands, factors, simplifies, and evaluates expressions. Also shows derivative and integral.

Input: `Expr= (x+3)^2−9`

PRACTICE QUESTION

Which of the following is equivalent to $(2x + 3)^2 - 4x^2$?

Enter: `(2x+3)^2−4x^2`

Answer: 12x + 9

Find y = mx + b

Given slope m and y-intercept b, shows the line equation and x-intercept.

Inputs: `m= 2` `b= −3`

PRACTICE QUESTION

A line has slope 3 and passes through (0, −5). What is the x-intercept?

Enter: `m= 3, b= −5`

Answer: x-intercept = (5/3, 0) ≈ (1.667, 0)

Find Slope

Calculates slope, midpoint, distance, and line equation from two points.

Inputs: `x1= 1` `y1= 2` `x2= 4` `y2= 8`

PRACTICE QUESTION

Line p passes through (–3, 7) and (5, 1). What is the slope of p?

Enter: x1= –3 y1= 7 x2= 5 y2= 1

Answer: m = –3/4

Point-Slope Form

Given a point and slope, produces point-slope and slope-intercept forms.

Inputs: `x1= 2` `y1= 5` `m= 3`

PRACTICE QUESTION

Write the equation of the line through (4, –1) with slope –2 in slope-intercept form.

Answer: y = –2x + 7

Parallel or Perpendicular?

Tests whether two lines (given their slopes) are parallel, perpendicular, or neither.

Inputs: `m1= 3` `m2= -1/3`

PRACTICE QUESTION

Line a has slope 4. Line b has slope –1/4. Are they parallel, perpendicular, or neither?

Answer: Perpendicular (4 × –1/4 = –1)

Find Parallel/Perpendicular Lines

Given a slope and a point, finds the equations of the parallel and perpendicular lines through that point.

Inputs: `m= 2` `x1= 1` `y1= 3`

PRACTICE QUESTION

Find the equation of the line perpendicular to y = 3x + 1 that passes through (6, 2).

Enter: m= 3 x1= 6 y1= 2

Answer: y = –1/3 x + 4

📖 Reference Cards in Algebra

PEMDAS — Order of operations reminder
Direct Variation (y = kx) — k is the constant, line through origin
Linear Functions — y = mx + b, point-slope, standard form, parallel/perp rules
Absolute Value — |x| definition, y = a|x − h| + k vertex form
Intervals — (a,b), [a,b], half-open, infinity notation

2. Advanced Math (35% of SAT Math)

Quadratics, polynomials, rational/radical expressions, exponents, and logarithms.

Quadratic Equation ($ax^2 + bx + c = 0$)

Applies the quadratic formula. Shows discriminant, solutions (real or complex), and vertex.

Inputs: `a= 1` `b= −5` `c= 6`

PRACTICE QUESTION

What are the solutions to $x^2 - 7x + 12 = 0$?

Enter: a=1 b=−7 c=12

Answer: $x = 3$, $x = 4$

Complete the Square

Converts $ax^2 + bx + c$ to vertex form $a(x - h)^2 + k$.

Inputs: `a= 1` `b= 6` `c= 5`

PRACTICE QUESTION

Rewrite $x^2 + 8x + 12$ in vertex form.

Enter: a=1 b=8 c=12

Answer: $(x + 4)^2 - 4$, vertex at $(-4, -4)$

Complete Sq → Vertex

Finds vertex, axis of symmetry, and direction of opening from a, b, c.

Inputs: `a= −2` `b= 8` `c= −3`

PRACTICE QUESTION

The function $f(x) = -x^2 + 6x - 5$ has a maximum value. What is it?

Enter: a=−1 b=6 c=−5

Answer: Maximum = 4 at vertex $(3, 4)$

Polynomial Explorer

Finds degree, zeros, factored form, derivative, and coefficients of a polynomial.

Input: `p(x)= x^3−6x^2+11x−6`

PRACTICE QUESTION

How many real zeros does $p(x) = x^3 - 4x$ have?

Enter: x^3−4x

Answer: 3 zeros: $x = -2, 0, 2$

Rational Expression Simplifier

Factors numerator and denominator, simplifies, and shows domain restrictions.

Inputs: `Numerator= x^2−4` `Denominator= x+2`

PRACTICE QUESTION

Simplify: $(x^2 - 9) / (x^2 - x - 6)$

Enter: Num = x^2−9, Den = x^2−x−6

Answer: $(x + 3)/(x + 2)$, $x \neq 3$, $x \neq -2$

Radical Simplifier

Simplifies radical expressions and provides exact and decimal results.

Input: `Expression= sqrt(72)`

PRACTICE QUESTION

Simplify $\sqrt{200} + \sqrt{50}$.

Enter: `sqrt(200)+sqrt(50)`

Answer: $15\sqrt{2} \approx 21.213$

Factor Integers

Prime factorization, all divisors, divisor pairs, and primality test.

Input: `Integer= 360`

PRACTICE QUESTION

How many positive divisors does 120 have?

Enter: `120`

Answer: 16 divisors. $120 = 2^3 \times 3 \times 5$

Powers

Computes b^e, its inverse, and the n-th root.

Inputs: `Base= 5` `Exponent= 3`

PRACTICE QUESTION

If 2^x = 64, what is x?

Enter: `Base=2 Exponent=6` → see that $2^6 = 64$

Answer: x = 6

GCD & LCM

Finds GCD and LCM with Euclidean algorithm steps and prime factorizations.

Inputs: `Integer1= 24` `Integer2= 36`

PRACTICE QUESTION

Two runners lap a track every 12 and 18 minutes. When do they next meet at start?

Enter: 12 and 18

Answer: LCM(12, 18) = 36 minutes

Solve Exponential/Log Equation

Solves equations involving exponents or logarithms for x.

Input: `Equation= 3^x=81`

PRACTICE QUESTION

Solve: 5^(2x) = 125

Enter: `5^(2x)=125`

Answer: x = 3/2

Rule of 72

Estimates how long it takes an investment to double at a given interest rate.

Input: `Rate(%)= 8`

PRACTICE QUESTION

An account earns 6% annual interest. Approximately how many years to double?

Answer: 72 ÷ 6 = 12 years

Exponential Growth

Computes $P(t) = P_0(1 + r)^t$ and continuous growth $P_0e^{(rt)}$.

Inputs: `P0= 1000` `rate(%)= 5` `t= 10`

PRACTICE QUESTION

A population of 500 bacteria grows 20% per hour. How many after 5 hours?

Enter: `P0=500 rate=20 t=5`

Answer: ≈ 1,244

Evaluate Logarithm

Computes log_b(value) for any base.

Inputs: `Base= 2` `Value= 32`

PRACTICE QUESTION

What is $\log_3(81)$?

Enter: `Base=3 Value=81`

Answer: 4 (since $3^4 = 81$)

Change of Base

Converts log_b(x) to ln(x)/ln(b).

Inputs: `Base= 5` `Value= 30`

PRACTICE QUESTION

Express $\log_7(50)$ as a ratio of natural logarithms and evaluate.

Answer: ln(50)/ln(7) ≈ 2.012

📖 Reference Cards in Advanced Math

Exponent Rules — $x^a \cdot x^b = x^{(a+b)}$, $(x^a)^b = x^{(ab)}$, $x^0 = 1$, $x^{(-a)} = 1/x^a$

Log Rules — $\log(xy) = \log x + \log y$, $\log(x^n) = n \cdot \log x$, change of base

3. Functions

Function Explorer

Finds zeros, y-intercept, symmetry (even/odd), critical points, min/max, derivative, and integral.

Input: `f(x)= x^2−4x+3`

PRACTICE QUESTION

The function $f(x) = x^2 - 6x + 8$ has how many x-intercepts?

Enter: x^2−6x+8

Answer: 2 zeros at x = 2 and x = 4

Evaluate Function

Plugs a value into f(x) to compute f(a).

Inputs: `f(x)= 3x^2−2x+1` `x= 4`

PRACTICE QUESTION

If $f(x) = 2x^3 - x + 5$, what is $f(-2)$?

Enter: f(x)=2x^3−x+5 x=−2

Answer: f(−2) = −9

Intersection of Two Functions

Finds where f(x) = g(x) — both exact and numeric solutions.

Inputs: `f(x)= x^2` `g(x)= 2x+3`

PRACTICE QUESTION

At what point(s) does $y = x^2$ intersect $y = x + 6$?

Enter: f(x)=x^2 g(x)=x+6

Answer: x = 3 (y = 9) and x = −2 (y = 4)

Function Arithmetic (f+g, f–g, fg, f/g)

Computes combined function operations.

Inputs: `f(x)= x^2+1` `g(x)= x−3`

PRACTICE QUESTION

If f(x) = x + 2 and g(x) = x^2, what is (f · g)(3)?

Answer: (f·g)(x) = $x^3 + 2x^2$, so (f·g)(3) = 27 + 18 = 45

Composition f(g(x))

Computes f(g(x)) and g(f(x)).

Inputs: `f(x)= x^2` `g(x)= x+3`

PRACTICE QUESTION

If f(x) = 2x + 1 and g(x) = x^2, what is f(g(3))?

Enter: `f(x)=2x+1 g(x)=x^2`

Answer: g(3) = 9, f(9) = 19

Complex Numbers (1 or 2)

Real/imaginary parts, modulus, angle, conjugate. For 2 numbers: +, −, ×, ÷.

Input: `z1= 3+4i`

PRACTICE QUESTION

What is (2 + 3i)(4 − i)?

Enter two complex: `z1=2+3i z2=4−i`

Answer: 11 + 10i

4. Problem Solving & Data Analysis (15% of SAT Math)

Ratios, percentages, statistics, probability, and data interpretation.

Solve Proportion

Cross-multiplication: a/b = c/d. Leave one blank to solve.

Inputs: `a=3` `b=4` `c=9` `d=` (leave blank)

PRACTICE QUESTION

If 5/8 = x/24, what is x?

Enter: a=5 b=8 c= (blank) d=24

Answer: x = 15

Percent Change

Calculates (New – Old)/|Old| × 100 with step-by-step work and multiplier.

Inputs: `Old Value= 80` `New Value= 100`

PRACTICE QUESTION

A shirt originally costs $45 and is now $36. What is the percent decrease?

Enter: `Old=45 New=36`

Answer: 20% decrease

Mean, Median, Mode

Computes central tendency measures from a comma-separated data list.

Input: `Data(csv)= 5,8,3,7,9,3,6`

PRACTICE QUESTION

The test scores are: 72, 85, 91, 68, 85, 77, 90. What is the median?

Enter: `72,85,91,68,85,77,90`

Answer: Median = 85

Standard Deviation

Population σ, sample s, and variance from comma-separated data.

Input: `Data(csv)= 4,8,6,5,3,7,9`

PRACTICE QUESTION

Data set: 10, 12, 14, 16, 18. What is the population standard deviation?

Enter: `10,12,14,16,18`

Answer: σ ≈ 2.828

Five-Number Summary

Min, Q1, Median, Q3, Max, and IQR.

Input: `Data(csv)= 2,4,6,8,10,12,14`

PRACTICE QUESTION

Data: 3, 5, 7, 9, 11, 13, 15, 17. What is the IQR?

Enter: `3,5,7,9,11,13,15,17`

Answer: Q1 = 6, Q3 = 14, IQR = 8

Linear Regression

Fits y = mx + b to paired data. Shows slope, intercept, and R^2.

Inputs: `x vals= 1,2,3,4,5` `y vals= 2,4,5,4,5`

PRACTICE QUESTION

Hours studied: 1, 2, 3, 4, 5. Scores: 50, 55, 65, 70, 80. Find the best-fit line.

Enter: x = `1,2,3,4,5` y = `50,55,65,70,80`

Answer: y = 7x + 43, R^2 ≈ 0.982

Two-Way Table

Builds a 2×2 table and computes joint, marginal, and conditional probabilities.

Inputs: `a(R1C1)=30` `b(R1C2)=20` `c(R2C1)=10` `d(R2C2)=40`

PRACTICE QUESTION

In a survey, 45 students like math, 25 don't. Of those who like math, 30 are in honors. Of those who don't, 5 are in honors. What is P(Honors | Likes Math)?

Enter: a=`30` b=15 c=5 d=`20`

Answer: P(Honors|Likes Math) = 30/45 = 0.667

Margin of Error

Computes margin of error and confidence interval for a proportion.

Inputs: `p= 0.6` `n= 200` `Conf Level%= 95`

PRACTICE QUESTION

In a poll of 400 voters, 55% support a candidate. What is the 95% confidence interval?

Enter: p=`0.55` n=`400` `Conf`=95

Answer: MOE ≈ 0.0488, CI: (50.1%, 59.9%)

Basic Probability

P = favorable/total, odds for, odds against, complement.

Inputs: `Favorable= 3` `Total= 10`

PRACTICE QUESTION

A bag has 4 red, 6 blue, and 5 green marbles. What is P(blue)?

Enter: `Favorable`=6 `Total`=15

Answer: P = 6/15 = 0.4 = 40%

nCr Combinations / nPr Permutations

C(n,r) for unordered selections; P(n,r) for ordered arrangements.

Inputs: `n= 10` `r= 3`

PRACTICE QUESTION

A committee of 3 is chosen from 8 people. How many ways?

Enter: n=8 r=3

Answer: C(8,3) = 56

5. Geometry & Trigonometry (15% of SAT Math)

Areas, volumes, circles, triangles, coordinate geometry, and right-triangle trig.

Circle

Area, circumference, and diameter from radius.

Input: `r= 7`

PRACTICE QUESTION

A circular garden has radius 10 feet. What is its area?

Answer: $A = \pi(10^2) = 100\pi \approx 314.159$ ft²

Circle Equation

Converts between standard form $(x-h)^2 + (y-k)^2 = r^2$ and general form. Shows center, radius, area.

Inputs: `h= 3` `k= −2` `r= 5`

PRACTICE QUESTION

A circle has center (−1, 4) and radius 3. What is the equation in standard form?

Enter: h=−1 k=4 r=3

Answer: $(x + 1)^2 + (y - 4)^2 = 9$

Pythagorean Theorem

Computes $c = \sqrt{(a^2 + b^2)}$.

Inputs: `a= 5` `b= 12`

PRACTICE QUESTION

A right triangle has legs 8 and 15. What is the hypotenuse?

Answer: $c = \sqrt{(64 + 225)} = \sqrt{289} = 17$

Triangle / Rectangle / Square / Parallelogram / Trapezoid

Area, perimeter, and diagonal for each shape.

PRACTICE QUESTION

A trapezoid has parallel sides 6 cm and 10 cm, height 4 cm. What is the area?

Use Trapezoid: a=6 b=10 h=4

Answer: A = ½(6 + 10)(4) = 32 cm²

Sphere / Cube / Cylinder / Cone / Pyramid / Rectangular Prism

Volume, surface area, and related measurements for 3D shapes.

PRACTICE QUESTION

A cone has radius 4 and height 9. What is its volume?

Use Cone: r=4 h=9

Answer: V = ⅓π(16)(9) = 48π ≈ 150.796

Pyramid Volume

V = (1/3)lwh — directly from the SAT reference sheet.

Inputs: `l= 6` `w= 4` `h= 9`

PRACTICE QUESTION

A rectangular pyramid has a base 8 × 5 and height 12. What is the volume?

Enter: l=8 w=5 h=12

Answer: V = (1/3)(8)(5)(12) = 160

Sector Area & Arc Length

Area = ½r²θ and arc length = rθ (angle in degrees, converted internally).

Inputs: `r= 10` `theta= 60`

PRACTICE QUESTION

A pizza slice from a 14-inch diameter pizza has a 45° central angle. What is the area of the slice?

Use Sector: r=7 theta=45

Answer: A = ½(49)(π/4) ≈ 19.242 in²

Distance Formula / Midpoint Formula

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ and midpoint = $((x_1+x_2)/2, (y_1+y_2)/2)$.

Inputs: `x1=1` `y1=2` `x2=7` `y2=10`

PRACTICE QUESTION

What is the distance between (–3, 4) and (5, –2)?

Enter: x1=–3 y1=4 x2=5 y2=–2

Answer: $d = \sqrt{(64 + 36)} = \sqrt{100} = 10$

Solve Right Triangle

Enter any 2 of 3 sides (a, b, hypotenuse c). Finds missing side, angles, area, perimeter.

Inputs: `a= 5` `b=` (blank) `c= 13`

PRACTICE QUESTION

A ladder leans against a wall. Base is 6 ft from the wall, ladder is 10 ft. How high up the wall?

Enter: a=6 b= (blank) c=10

Answer: b = 8 ft

Evaluate sin/cos/tan

Computes sin, cos, and tan for any angle in degrees.

Input: `x (deg)= 30`

PRACTICE QUESTION

What is the value of sin(45°)?

Enter: 45

Answer: $\sin(45°) = \sqrt{2}/2 \approx 0.707$

Degree ↔ Radian

Converts between degrees and radians.

Input: `Degrees= 180`

PRACTICE QUESTION

Convert 270° to radians.

Answer: $270° = 3\pi/2 \approx 4.712$ rad

📖 Reference Cards in Geometry & Trig

SOH-CAH-TOA — sin = O/H, cos = A/H, tan = O/A, plus $\sin^2 + \cos^2 = 1$

Circle Properties — $A = \pi r^2$, $C = 2\pi r$, equation $(x-h)^2 + (y-k)^2 = r^2$

Unit Circle — Key angle coordinates (0°, 30°, 45°, 60°, 90°, etc.)

Special Right Triangles — 45-45-90 ($x, x, x\sqrt{2}$) and 30-60-90 ($x, x\sqrt{3}, 2x$), Pythagorean triples

6. SAT Specials

Sigma Notation

Computes Σ f(n) from n = a to n = b.

Inputs: `f(n)= n^2` `from n= 1` `to n= 10`

PRACTICE QUESTION

What is the sum of 2n + 1 for n = 1 to 5?

Enter: `f(n)=2n+1 from=1 to=5`

Answer: 3 + 5 + 7 + 9 + 11 = 35

Count Integers / Count Divisible

Counts integers in a range, optionally divisible by N and/or M (uses inclusion-exclusion).

Count: `Low=1` `High=100` Divisible: + `N=3` `M=5`

PRACTICE QUESTION

How many integers from 1 to 200 are divisible by both 4 and 6?

Use Count Divisible: `Low=1 High=200 N=4 M=6`

Answer: LCM(4,6) = 12. Count = 16 integers divisible by both.

Integer Solutions: 2 or 3 Variables

Solves equations in multiple variables using CAS.

Input: `Equation= x+y=10`

PRACTICE QUESTION

Find all positive integer solutions to 3x + 2y = 20.

Answer: (2,7), (4,4), (6,1)

📖 SAT Reference Formulas

Given on SAT: $A = \pi r^2$, $C = 2\pi r$, $A = lw$, $A = \frac{1}{2}bh$, $a^2 + b^2 = c^2$, $V = lwh$, $V = \pi r^2 h$, $V = \frac{4}{3}\pi r^3$, $V = \frac{1}{3}\pi r^2 h$, $V = \frac{1}{3}lwh$
Must Memorize: $m = (y_2 - y_1)/(x_2 - x_1)$, $y = mx + b$, $Ax + By = C$, quadratic formula, vertex $h = -b/2a$, circle equation, SOH-CAH-TOA

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